Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Multi-Agent AI · Distributed Systems · Data Platform · GenAI Analytics · Python · AWS · LangGraph · Workflow Orchestration · 9yr Production

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PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience in distributed systems, data platforms, and multi-agent AI - the exact combination AWS Analytics Engineering is scaling. LeaseI is the direct proof of the multi-agent AI pattern: an 8-agent LangGraph StateGraph implementing workflow orchestration end-to-end with conditional routing, tool integrations, and deterministic guardrails - achieving 93.6% precision, 87.9% F1 across 1,240 test cases. AgentProbe demonstrates GenAI analytics at the platform level: a CI-integrated evaluation engine measuring correctness, groundedness, hallucination rate, latency, and token efficiency across multi-provider LLM pipelines, with structured tracing across all agent steps and batch evaluation pipelines for offline analysis. At Entrata I owned mission-critical Python distributed systems for 7 years at 99.5% uptime - RabbitMQ async event-driven processing with retry logic and idempotency guarantees, 200+ SQL optimizations across petabyte-scale operational data, Docker microservices, GitHub Actions CI/CD, and AWS S3/EC2/RDS/Glue infrastructure. I wrote design documents for 20+ production modules, led projects spanning multiple engineers, and mentored 5+ engineers - the technical leadership and operational excellence AWS expects at Senior SDE. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Multi-Agent AI & Orchestration: Python (9yr), LangGraph, LangChain, Multi-Agent Orchestration, Workflow Orchestration, Agentic AI, Autonomous Agents, Tool Calling, HITL Guardrails, Agent Frameworks, MCP (Model Context Protocol), Anthropic API, OpenAI API, Ollama, Prompt Engineering, Output Validation, Confidence Scoring, GenAI Analytics, AI Evaluation, LLM-as-Judge, Hallucination Detection, Groundedness, pass@k Harness

Distributed Systems & Data Platform: Distributed Systems, Event-Driven Architecture, RabbitMQ (async processing, retry logic, idempotency, dead-letter handling), Microservices, FastAPI, REST APIs, PostgreSQL, MySQL, Oracle, Redis, ChromaDB, Vector Databases, RAG Pipelines, Data Pipelines, Schema Design, Query Optimization, Indexing, Partitioning, High-Throughput Systems, Reliability Engineering

AWS, Observability & Engineering Excellence: AWS (S3, EC2, RDS, Glue, Lambda), Docker, GitHub Actions CI/CD, Structured Logging, Distributed Tracing, Observability, Monitoring, Alerting, Incident Response, Root-Cause Analysis, Runbooks, Automated Testing, Regression Testing, Design Documents, Technical Leadership, Code Reviews, Mentoring, Agile/Scrum, Linux, Bash, Git

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Multi-Agent Workflow Delivery

- Led 0→1 delivery of a production multi-agent workflow system (LangGraph, FastAPI, Python) as sole technical owner - designed orchestration architecture, implemented workflow state management and conditional routing, built evaluation pipeline and CI validation gate, delivered tracing and observability infrastructure for ongoing operation. Measured outcomes drove iteration decisions throughout.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977) on AWS-compatible cloud infrastructure; designed observability dashboards improving research comprehension 40%; delivered technical output across 3 research pipelines with cross-functional stakeholder reporting; mentored 600+ students in Python, SQL, and distributed systems.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Distributed systems & data platform at production scale** - owned mission-critical Python backend distributed systems for 7 years at 99.5% uptime serving 100K+ housing units; RabbitMQ event-driven async processing orchestrating thousands of daily workflows with retry logic, idempotency guarantees, and dead-letter handling; 200+ PostgreSQL/MySQL/Oracle query optimizations through execution plan analysis, indexing, and partitioning (30% performance gain); AWS S3/EC2/RDS/Glue infrastructure; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); GitHub Actions CI/CD with automated regression testing.
- **Operational excellence & observability** - built structured logging, monitoring, and alerting infrastructure; authored runbooks for all owned systems; led incident response with RCA across production outages; built internal debugging tools reducing engineering triage time 40%; kept bug count below 10 across 20+ production modules for 6+ consecutive months over 120+ Agile sprints - measurable reliability improvement at team scale. Python/Bash automation eliminating 80% of manual operational workload.
- **Technical leadership & engineering excellence** - wrote design documents and architecture decisions for 20+ production modules; led projects spanning multiple engineers from concept through deployment; mentored 5+ engineers on distributed systems design and production best practices; 300+ peer code reviews; delivered technical guidance to product, compliance, and operations stakeholders; 100% PHI compliance on AWS S3 data migration.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

Leasel: Multi-Agent Orchestration Framework - Production Workflow Automation

Multi-Agent · LangGraph · Workflow Orchestration · Tool Integration · Measurable Outcomes

Python · LangGraph (8-agent) · FastAPI · RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production multi-agent orchestration framework - the exact pattern AWS Analytics Engineering is building for workflow automation at scale. **Orchestration architecture:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) implementing complex multi-step workflow automation with conditional routing, dynamic task delegation, and tool integrations - the same pattern that scales to thousands of daily analytical workflows. **Reusable components:** modular agent APIs and shared state interfaces enable downstream teams to integrate specialized capabilities without rebuilding orchestration logic. **Operational excellence:** deterministic HITL guardrails enforce safe execution; structured output validation at every agent boundary; confidence-based routing avoids unnecessary LLM calls. **Measured outcomes:** 93.6% precision, 87.9% F1 across 1,240 test cases.

AgentProbe: GenAI Analytics & Evaluation Platform - Data-Driven AI Decision Making

GenAI Analytics · Evaluation · Observability · CI Gate · Platform · Open Source

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production GenAI analytics and evaluation platform - surfacing data-driven insights on AI system quality to drive leadership decisions, the same pattern AAE delivers for AWS product leadership. **Analytics engine:** 5-metric quality measurement (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring; pass@k multi-trial harness quantifying variance in non-deterministic outputs. **Platform services:** 15-endpoint FastAPI backend with structured data services for evaluation runs, metric aggregation, and multi-provider LLM routing; offline batch evaluation pipelines for large-scale analysis. **Observability:** distributed tracing across all agent steps; monitoring and alerting on quality drift; feedback loops from production into evaluation suites. **CI integration:** GitHub Actions regression gate - zero quality regressions across 1,240+ test cases. Open-source, multi-team deployable.

AI on FHIR: Full-Stack Backend - AWS Infrastructure & Data Services

AWS · Distributed Backend · REST APIs · PostgreSQL · Compliance · Production

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack production backend on AWS infrastructure - LLM-powered tool invocation translating natural-language queries into structured API calls against regulated healthcare data; AWS Lambda serverless with API gateway; PostgreSQL data services with HIPAA-compliant RBAC, full data isolation, and audit trail; Docker containerization; complete observability stack. Demonstrates distributed backend reliability engineering, AWS infrastructure management, and production operational excellence in a regulated environment - ~60% reduction in manual retrieval time.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017